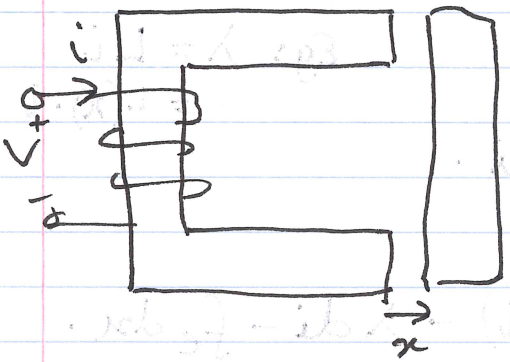
Lecture 19.

Last time: * How do we calculate force in electromechanical system?

So, we used energy (1) W_m calculate. x = constant.



$$dW_m = \int i\, d\lambda \quad @\ x = \text{const.}$$

$$= \int_0^{\lambda} \frac{\lambda'}{L(x)}\, d\lambda' = \frac{\lambda^2}{L(x)}.$$

(2) Calculate, $f_e = -\left.\frac{\partial W_m}{\partial x}\right|_{\lambda=\text{const}}$

$$= -\lambda^2 \frac{\partial}{\partial x}\left[\frac{1}{L(x)}\right].$$

$$\propto -\lambda^2 \frac{\partial}{\partial x}\left[R(x)\right]$$

Bottomline, the direction of force is such that it minimizes reluctance or maximizes inductance.

(3) Substitute $\lambda = L(x) \cdot i$.

$$f_e = -\left\{L(x)\cdot i\right\}^2 \frac{\partial}{\partial x}\left[\frac{1}{L(x)}\right].$$

Two challenges: ① Keeping λ constant while doing spatial derivative.
- mathematically okay but intuitively difficult to digest.

② Inversion of inductance.
- okay for single-excited system.
- later on we will see multiple coils can exist. Good luck with inversion.

$\omega_m = (\lambda, x, i)$

Q. Is there an alternative method to calculate force of electric origin?

So, far we considered, $\omega_m = f(\lambda, x)$.

$$d\omega_m = i\,d\lambda - f_e\,dx$$

Goal: Get rid of λ as an independent variable.

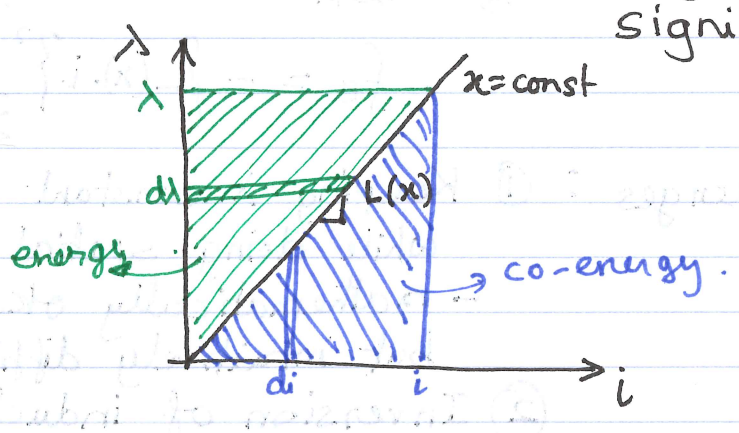
But, $\lambda = g(i, x)$ Eg: $\lambda = Li$
$= L(x)\cdot i$.

$\therefore\ d(\lambda i) = i\,d\lambda + \lambda\,di$.

$\therefore\ d\omega_m = d(\lambda i) - \lambda\,di - f_e\,dx$.

or, $d\underbrace{(\lambda i - \omega_m)}_{\omega_m'} = \lambda\,di + f_e\,dx$.

$\omega_m' = \lambda i - \omega_m$
$=$ co energy. (mathematical ~~concept~~ definition; no physical significance).



$\omega_m' = f(i, x)$ $\therefore\ d\omega_m' = \left.\frac{\partial \omega_m'}{\partial i}\right|_{x=\text{const}} di + \left.\frac{\partial \omega_m'}{\partial x}\right|_{i=\text{const}} dx$

So, this time first calculate,

Coenergy, $W_m' = \int_{i} \lambda \, di \Big|$ while x = constant

$= \int_0 L(x) \cdot i' \, di'$

$= \frac{1}{2} L(x) \cdot i^2$.

Next, $f_e = \frac{\partial}{\partial x} W_m' \Big|_{i=\text{const}}$.

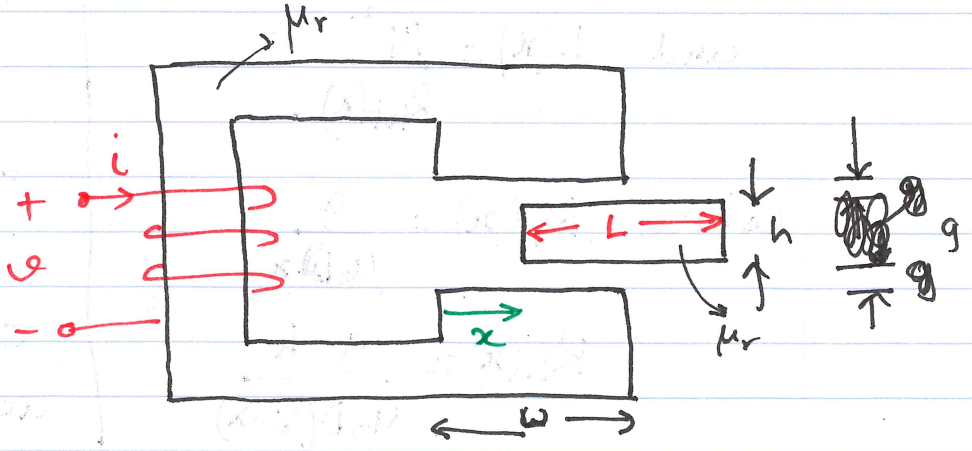
note no negative sign.

$\therefore$ $\boxed{f_e = \frac{1}{2} i^2 \cdot \frac{\partial}{\partial x}(L(x))}$ → we directly get f_e as a function of i and x (physically measurable).

↑ again f_e acts in the direction of increasing inductance.

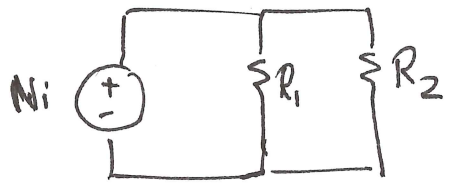
Example:

* No fringing
* $\mu_r \to \infty$
* depth : D.



Q. How should the plunger move?

$$\frac{1}{R_{tot}} = \frac{1}{R_1(x)} + \frac{1}{R_2(x)}.$$



$$R_1(x) = \frac{g}{\mu_0 A} = \frac{g}{\mu_0 D x}.$$

$$R_2(x) = \frac{g-h}{\mu_0 (w-x) D}.$$

Energy method:

Calculate energy,

$$W_m = \int i \, d\lambda \quad x = \text{const}.$$

$$= \frac{\lambda^2}{2L(x)}.$$

Calculate force, (λ = const)

$$f_e = -\frac{\lambda^2}{2} \cdot \left[\frac{\partial}{\partial x}\left[\frac{1}{L(x)}\right]\right]$$

$$= -\frac{L(x)^2 i^2}{2N^2} \cdot \frac{\partial}{\partial x}\left\{R_{tot}(x)\right\}$$

where $R_{tot}(x) = \dfrac{R_1(x) R_2(x)}{R_1(x) + R_2(x)}$

and $L(x) = \dfrac{N^2}{R_{tot}(x)}$.

where $R_1(x) = \dfrac{g}{\mu_0 D x}$

$$R_2(x) = \frac{g-h}{\mu_0 D (w-x)}$$

Coenergy method.

Calculate co-energy,

$$W_m' = \int \lambda \, di \quad x = \text{const}.$$

$$= \frac{1}{2} L(x) \cdot i^2$$

Calculate force, (i = const).

$$f_e = \frac{i^2}{2}\left[\frac{\partial}{\partial x}\left[\frac{N^2}{R_{tot}(x)}\right]\right]$$

$$= \frac{N^2 i^2}{2}\left[\frac{\partial}{\partial x}\left\{\frac{\mu_0 D x}{g} + \mu_0 \frac{(w-x) D}{g-h}\right\}\right]$$

$$= \frac{\mu_0 D N^2 i^2}{2}\left[\frac{1}{g} - \frac{1}{g-h}\right]$$

$$= -\frac{\mu_0 D N^2 i^2 h}{g(g-h)}.$$

constant force acting opposite to the direction of x

valid $0 < x < w$ and $L \geq w$.